

eyes to see, a nose to smell, fingers to feel and a brain to interpret it all, sensors are what add feelings to otherwise cold mechanical systems.

Without them, elevator doors would squish children on a daily basis, robotic arms would knock over people in factories and disaster would be commonplace in general. As technology gets more pervasive in our everyday life, and yet we want it all to stay as invisible as possible, it is sensors that will rule the future. They will make sure you stay warm or cool wherever you are, keep you safe, and be neither seen nor felt.

New types of sensors are always being developed, and one of the newest type are Electric Field (EF) sensors. These sensors work by sensing changes in electric fields between two fixed points, but are able to do so beyond barriers. This ability to sense movement and changes of electric fields through solid material makes them very promising for the future for a few reasons. Firstly, they can be hidden better

low-power, which also makes them ideal for all of our handheld devices, and since they don't need to be exposed in any way, they will also last longer – unlike say the IR sensor on a gadget, or the light sensor that adjusts your screen brightness.

One of the instances of application shown a few years ago by Intel was a robotic arm that used EF sensors to identify empty bottles. If three identical bottles of water were placed in front of the robot, only one of which was empty, without even touching any of the bottles, the robot was able to identify the empty bottle and knock it over. What this means is that we already have the ability to get a robot to know the difference between full and empty, which means the difference between light and heavy, which in turn can be translated into how much force to squeeze with in order to lift without slipping, and also without crumpling. The way you hold and lift a paper cup with water and without are different, and now robots can do that.



Programming your own prosthetic? Why not?

behind layers of solid material. They can be placed in a car's steering wheel, for instance, and will be able to detect when your hands move away from the wheel.

Software could be used to program variables – such as travelling above the speed of 25 kmph with no hands on the steering wheel – to sound alarms; just in case you've fallen asleep. It could also be programmed to slow down the car automatically, turn on the hazard lights, and bring the car to a gradual halt – in case the driver's had a heart attack, or has passed out. Secondly, these sensors are really

More recently, scientists at the Universidad Politécnica de Madrid, Spain showed off the *Rosphere*, which is basically a robot that moves about the way hamsters inside plastic balls do – by rolling. This robot will be equipped with cameras and EF sensors in order to be able to move about farms and to detect whether the soil has the appropriate moisture and temperature, and then report that back to a receiver that the farmer owns. This will make farming easier, and will also amount to better, healthier yields of crops, as farmers will know exactly what's

Medical advancements



Medical tech has advanced so far beyond what it was just a decade ago that out of touch doctors will be shell-shocked

by the advancements of today. Here is a bundle of links that serve as examples of the advancement that sensors and networks have brought about in medical technology. <http://dgigit.in/CSJuly7>

happening in even small patches of farm land – even knowing that a particular small area is drier than others, and thus changing the sprinkler settings to water that bit more would result in a better overall crop, for example.

Engines and transport

The world is a smaller place thanks to all the various types of engines that mankind has developed. Jet engines take you across continents and internal combustion engines across cities and countries even. Rockets take us to outer space and the moon, but that's where we've stopped. Engines help us travel, transport goods and allow us to manage supply and demand across the globe. However, it's these very engines that are also burning up fuel and causing environmental damage.

Electric engines try and reduce emissions in the effort to go green, but even electricity is generated using brown methods (the opposite of green). We're sort of at an impasse now, with fuel efficiency maxing out for terrestrial engines and space travel suspended because of lack of interstellar drives – we'd settle for inter-planetary drives, actually.

So where do we go from here? Thanks to technology, we still have a thin sliver of a silver lining to look forward to. Even modifications to existing engines could result in greener variations, and if not that, at least result in more power for less fuel – given the skyrocketing fuel prices, this is certainly a much needed solution.

Solar sails

Using our sun as an engine is perhaps the cheapest and best way to travel the vast distances between planets. Solar sails